

CHAPTER

5

ENHANCED VIBRATION DIAGNOSTICS USING CASCADE DIAGRAMS

5.0 INTRODUCTION

Cascade diagrams (sometimes referred to as "Waterfall Plots" or "Spectral Maps"), give the analyst a view of the machine vibration characteristics relative to operating speed or the actual running time. The cascade plot graphically displays resonant events, rotor rubs, oil whirl, oil whip, beat frequency events and other mechanical faults in a series of frequency spectra which are generated as the machine changes speed, or as time progresses. Signals that are changing slowly with respect to time are difficult to see in real time. However, periodic sampling with waterfall plots can clearly show these changes which might otherwise have gone unnoticed.

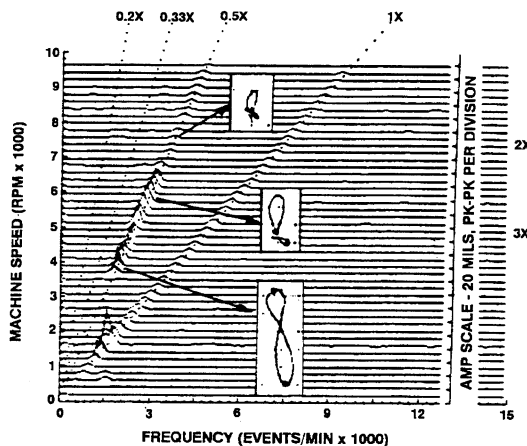
For example, in the synthetic fibers industry which employs many winders (for winding small strands of yarn filament onto bobbins up to a package size of approximately 25 to 30 lbs), these machines often require anywhere from 30 or 40 minutes up to about 8 or 9 hours to wind a full package (see drawing of such a winder in Figure 8), these machines are forced to pass through several resonances which can often be quite severe. By examining waterfall plots captured throughout the package build, one can often see when the machine passes through resonant events, or when it encounters beat frequencies increasing amplitude for a few minutes (when either the speed or a running speed harmonic of one component closely approaches the operating speed of another component). If this entire package build (known as a "doff cycle") is captured on tape, or within the memory buffer of a spectrum analyzer, this will allow one to closely examine these resonant and beat frequency events, even to the point of calculating the resonant amplitude amplification factor (Q), as well as the damping factor (ζ). This will be covered in later portions of this chapter and portions of other chapters in the text.

5.1 DIAGNOSIS OF ROTOR RUB PROBLEMS

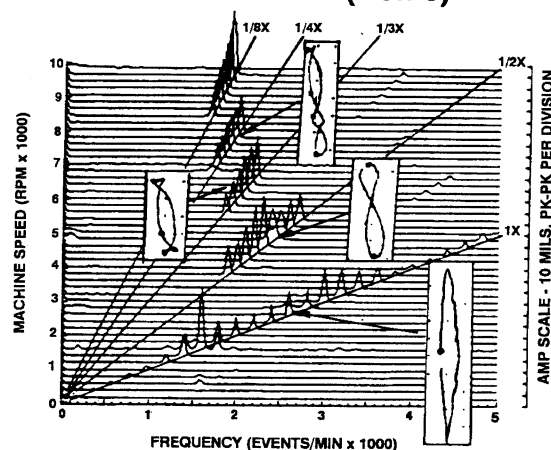
Cascade plots are especially helpful in diagnosing radial rubs. An example cascade plot is shown in Figure 1 which illustrates a partial rub. This cascade plot shows how vibration amplitude and frequency changes with machine RPM. It shows a series of single spectra in an amplitude versus frequency plot captured as the machine is brought up to speed. Note that the 1X RPM peak goes through a natural frequency at about 1500 to 1800 RPM. Later, the amplitude at $\frac{1}{2}$ X RPM goes through the same natural frequency, at which point the amplitude of the $\frac{1}{2}$ X RPM peak is much greater than the peak at 1X RPM when the speed of the machine is approximately 3000 to 3600 RPM. An unusual occurrence like this often indicates a partial radial rub. In these cases, one will have to ensure the subsynchronous frequency is .50X RPM (and not 40% to 48% of RPM which would point to another problem source - oil whirl).

Shaft orbit displays (known as "Lissajous Orbits") are also shown in Figure 1 and are likewise very helpful in diagnosing rubs. These orbit displays show the actual path the shaft itself follows inside the bearing (using the output from two eddy current proximity probes placed in the machine housing about 40 to 60 mils away from the shaft target, and with the probes located 90° apart from one another). The shape of the orbit is helpful in determining phase, as well as in distinguishing the nature of the rub, its severity, and probable frequency content. Lissajous orbits will be discussed separately in later sections of the seminar text.

Partial rubs will generate noticeable subsynchronous vibration which is clearly captured within a cascade diagram, but very likely will be missed if just capturing a "Peak Hold Spectrum", or just observing the display of a spectrum analyzer in real time. For this reason, cascade diagrams should always be captured on critical machinery when conducting acceptance tests on new or rebuilt machinery. If the subsynchronous vibration is caused by a partial rub, it will normally show integer subharmonics ($1/2X$ or $1/3X$ or $1/4X$ RPM, etc.). Several different integer subharmonics can simultaneously occur when a light rub with a lower radial force occurs. Such a response is shown in Figure 2 which features both the spectral and the orbit responses for $1X$, $1/2X$, $1/3X$, $1/4X$, and $1/5X$ RPM. In this case, note that maximum vibration occurred at $1X$ RPM and each of its 5 subharmonics when they accelerated and passed through the fundamental bending mode (first critical speed) in the vicinity of 1700 to 2100 CPM (28.3 - 35.0 Hz).



**FIGURE 1
CASCADE DIAGRAM INDICATING A PARTIAL RUB EVENT UNDER
HIGH FORCE (Ref. 3)**



**FIGURE 2
PARTIAL RUB EVENT WITH LOW RUB FORCE GENERATING MANY
INTEGER SUBHARMONICS (Ref. 3)**

Partial rub events can eventually lengthen in duration and impact finally leading to a full annular rub in which case a virtually continuous rub occurs. Figure 3 shows a cascade diagram taken during a full annular rub of a seal rubbing on a shaft. The shaft orbit displays are also included. Note the very "messy" orbit shown on the lower right of the plot which showed very unstable shaft motion within the bearing and seal. This plot shows how the response changed as the machine

was brought up to speed. Note that at lower speeds, little vibration was noticed. At higher speeds approaching 3500 RPM, a response at about 7000 CPM began (at that instant when 7000 CPM equalled 2X RPM). It remained at this same 7000 CPM frequency, even when the machine was brought all the way up to 4500 RPM. This 7000 CPM response corresponded to the rotor/seal system's natural frequency which was higher than the natural frequency of the rotor alone before it contacted the seal due to the additional stiffness supplied by the seal during the rub event. This 7000 CPM frequency, then, was the "speed" of the backward precession (known as "reverse precession"). This phenomenon will be discussed in the chapter covering the "Vibration Diagnostic Chart" under the "Rotor Rub" section.

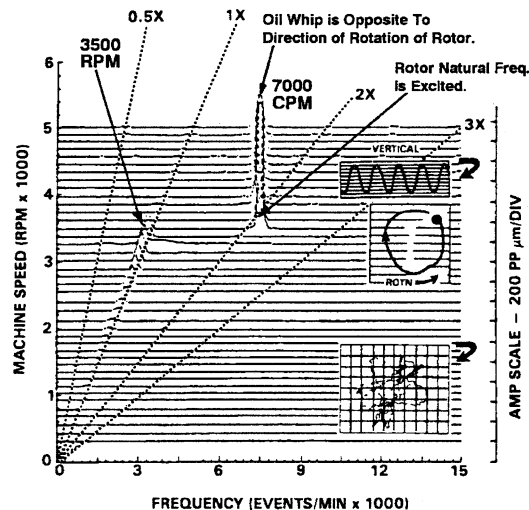


FIGURE 3
MULTIPLE IMPACTING SEAL RUB WHICH GENERATED REVERSE PRECESSION AT APPROXIMATELY 3500 RPM EXCITING A 7000 CPM ROTOR/SEAL SYSTEM NATURAL FREQUENCY (Ref. 3)

Figure 4 shows another type of full annular rub which is dominated by the response at 1X RPM followed by lower amplitude harmonics of running speed. In this case, the full annular rub was dominated by forward precession synchronous speed, and began experiencing subsynchronous vibration when the rotor accelerated to approximately 9000 RPM.

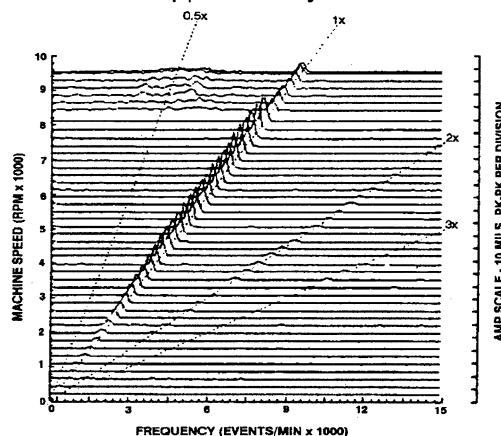


FIGURE 4
FULL ANNULAR RUB DOMINATED BY FORWARD PRECESSION SYNCHRONOUS SPEED (Ref. 2)

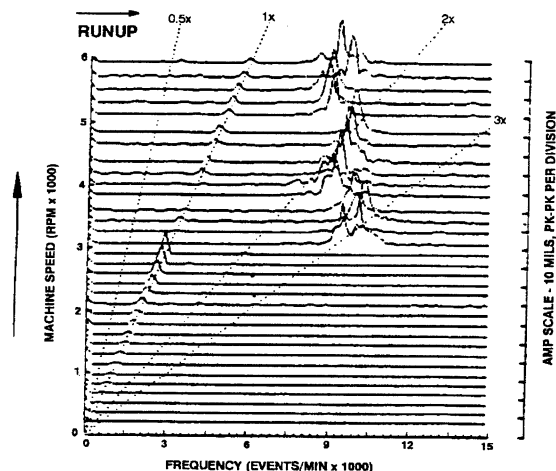


FIGURE 5
FULL ANNULAR RUB DURING RUNUP WITH SELF-EXCITED
BACKWARD PRECESSION (Ref. 2)

Figure 5 is an example of a full annular rub where a self-excited backward precession has occurred at a speed just below the first critical of the rotor. Note that even though the operating speed frequency (1X RPM) kept increasing, a wideband response of energy remained in the vicinity of 9000 CPM - a clear indication of a potentially severe rub event. However, Figure 5 clearly shows this could only be seen by examining a cascade diagram within a spectrum analyzer (or alternatively sent to a plotter from analyzer memory).

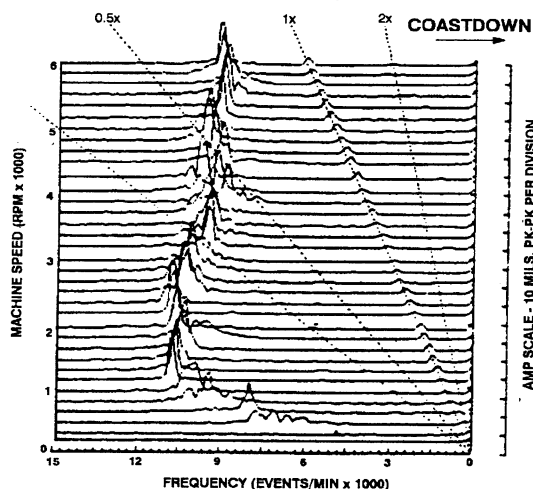


FIGURE 6
FULL ANNULAR RUB DURING COASTDOWN
(NOTE CONTINUED BACKWARDS PRECESSION UNTIL VERY LOW SPEED)
(Ref. 2)

Interestingly, note that the backward precession remained until the same rotor nearly came to a stop even after the unit was shut off and coasted down as shown by Figure 6. Once again, this was only clearly observed with a cascade diagram.

Note that, in both cases, when the backward precession was occurring, the comparative amplitude at 1X RPM was very low. Since the friction remained throughout the rub, it introduces non-linearities that generate many harmonics of running speed as well.

5.2 DIAGNOSTICS OF SERIOUS OIL WHIRL AND OIL WHIP PROBLEMS

Very large turbomachinery and compressors outfitted with journal bearings are usually outfitted with proximity probes set 90° apart, usually 45° to the left and right of vertical (10:30 and 1:30 o'clock positions). In this way, these critical machines can be permanently monitored. Also, this will allow an analyst to capture both cascade diagrams and shaft orbit displays similar to that shown in Figure 7 during startups and coastdowns.

On machines that are equipped with hydrodynamic journal bearings, the most likely cause of subsynchronous instability is oil whirl (which will be discussed in the chapter covering the "Vibration Diagnostic Chart"). Figure 7 shows the development of oil whirl just after the shaft was brought up to speed. Note that the shaft went into oil whirl at about 1800 RPM and remained in whirl until approximately 4000 RPM. At this point, note that the shaft 1X RPM began entering resonance which caused enough force to actually suppress and overcome oil whirl (during oil whirl, the highly pressurized oil wedge actually drives the shaft around rather than the shaft itself driving the oil wedge as the shaft should). However, once the machine had passed through resonance, the shaft forces and resultant vibrations drop off substantially allowing the oil whirl frequency to reappear as shown in Figure 7 which is inherently unstable.

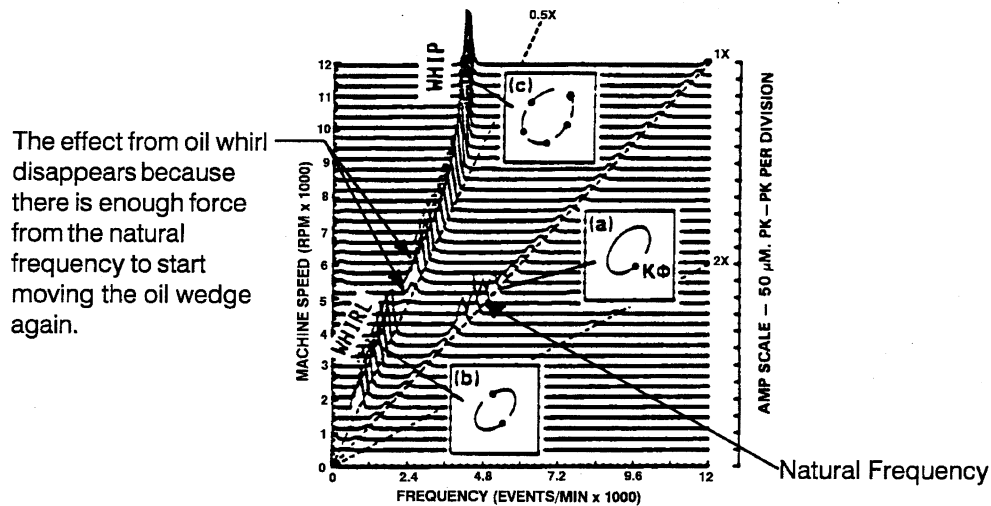


FIGURE 7
DEVELOPMENT OF OIL WHIRL JUST AFTER STARTUP
(FOLLOWED BY OIL WHIP FROM 9200-12,000 RPM) (Ref. 3)

When the rotor speed increased to just above 9200 RPM, it equalled about 2.2X its critical speed. At this time, the oil whirl which was approximately 45% of RPM was brought into coincidence with this critical speed. At this point, the oil whirl was suddenly replaced by oil whip. The oil whip frequency remained at the same 9200 CPM, independent of rotor RPM, even though the machine continued up to 12,000 RPM. In other words, the oil whirl frequency "locked into" the rotor natural frequency, thereby becoming oil whip (a highly unstable rotor problem). When a shaft goes into oil whip, its controlling dynamic factors become mass and stiffness in particular, with its resulting amplitude limited only by the bearing clearance. Note the high amplitudes at the 9200 CPM oil whip frequency in comparison to running speed (1X) in Figure 7. Left undetected and uncorrected, this serious instability can lead to catastrophic failure costing millions of dollars on these machines (not only in parts and manpower, but also in downtime). Very likely, only the use of cascade diagram capture could have detected (and properly diagnosed) this severe problem!

5.3 DIAGNOSTICS OF RESONANT FREQUENCIES

Cascade diagram acquisition is a powerful technique that enables an analyst to see what exactly is happening to his machine as it accelerates (or decelerates) from rest to operating speed. Such a plot can point directly to potential natural frequencies which must be avoided.

EXAMPLE A: DETECTION OF SYNTHETIC FIBER TEXTURING WINDER RESONANCE DURING OPERATION BY CAPTURING OF CASCADE DIAGRAMS

Figure 8 is a sketch of texturizing winder showing the vibration measurement points for the test and also the location of the major components of the winder. This constant surface-speed machine uses a friction roll (drive roll) to drive each of two bobbins held by what is called a chuck shaft (see Figure 8). When fully wound with yarn filament, each bobbin package normally weighs about 25 to 30 pounds.

NOTE: This Position was Chosen for Spectral Measurements During Package Builds.

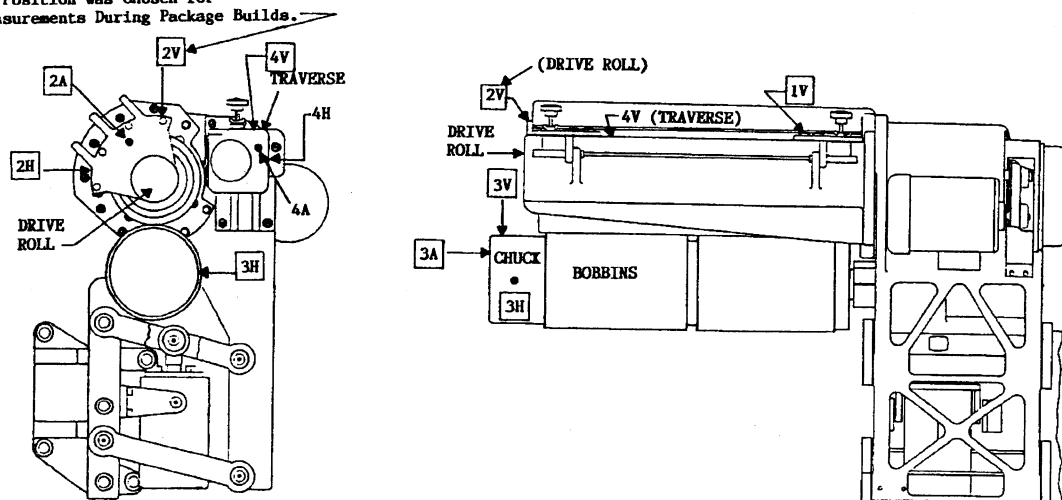


FIGURE 8
MEASUREMENT LOCATIONS FOR SYNTHETIC FIBERS WINDER

A controlled coastdown is shown in the cascade diagram of Figure 9 captured as the winder built a full 30 lb package on each bobbin. The friction roll remained at about 2670 RPM throughout the test. However, the chuck outfitted with the bobbins continually decreased in speed from 1620 to 855 RPM even though its surface speed remained constant (ft/min). Note the constant 2670 RPM friction roll speed in the diagram. But also note the first, second and third harmonic chuck speeds which all decelerated with increasing package size. Over on the right hand side are shown the chuck RPM, along with the vibration at 1X and 2X chuck RPM. Note the increase in amplitude as 2X chuck speed coincided with the friction roll speed of 2670 RPM at Point "A" on the diagram (chuck speed was 1330 - 1370 RPM in this region). The resultant beat frequency between 1X friction roll speed and 2X chuck speed caused an increase from .109 up to .372 in/sec, which was not unexpected.

After passing through this area, 2X chuck speed peak dropped back to .127 in/sec as shown in the plot. After this, vibration at 2X chuck speed was relatively low throughout the remainder of the package build. However, vibration at 1X chuck speed began increasing as the chuck speed decreased to 1035 RPM and peaked out at 885 RPM with an excessive amplitude of 1.79 in/sec as shown in the plot (and in the annotation just to the right of the plot). After passing through the apparent resonance, the amplitude at 1X chuck RPM

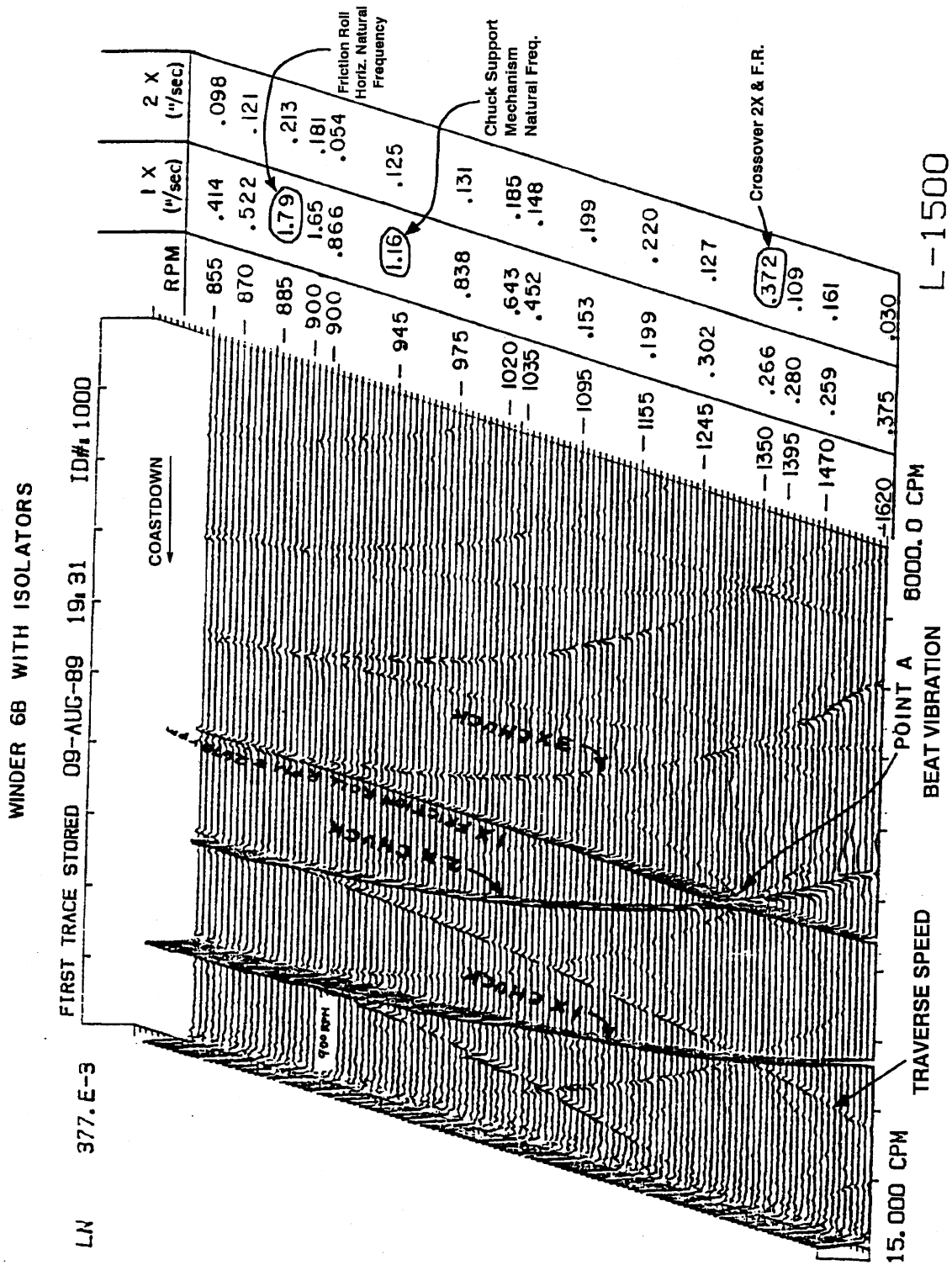


FIGURE 9

CASCADE DIAGRAM CAPTURED DURING A CONTROLLED COASTDOWN ON THE WINDER

decreased to .414 in/sec at 855 RPM as shown. After generating this cascade plot, impulse tests were performed on the chuck support mechanism and friction roll which indicated natural frequencies at approximately 990 CPM in the horizontal direction and 1050 CPM in the vertical direction. Also, the friction roll had a horizontal natural frequency at about 900 CPM. This indicates the reason for high vibration shown by 1X chuck RPM at speeds between 885 and 1020 RPM. This severe problem was vividly displayed by use of cascade diagrams. Here, the solution was application of damping materials rather than altering the chuck and friction roll natural frequencies since this was considered impractical and time consuming at this time for this variable speed machine (controlled by an inverter).

By referring to Figure 9, one can see the imminent coincidence of the natural frequency at about 900 CPM begin to occur when 1X chuck speed slowed to approximately 945 to 975 RPM. At other times, cascade plots taken during runups or coastdowns can reveal very strange behavior as shown below:

EXAMPLE B: DETECTION OF A GODET ROLL ASSEMBLY VIBRATION PROBLEMS FOR THIS SYNTHETIC FIBERS MACHINE USING CASCADE DIAGRAMS

Figure 10 is a sketch of a Godet Roll Assembly for a synthetic fiber operation which shows the various vibration measurement points as well as the identification of the rolls. Rolls 3A and 3B (each of which were heated) were supposed to be capable of operating up to 6000 RPM at a production rate of approximately 3000 meters/min (MPM). Unfortunately, when they tried to operate the panel at speeds just above 4000 RPM (2000 mpm), they encountered high vibration problems which resulted in product quality problems. The client desired Technical Associates to (1) determine the cause of such high vibration at operating speeds far below that rated for the machine; and (2) submit recommendations which would resolve this problem and truly allow them to operate the machine up to 6000 RPM.

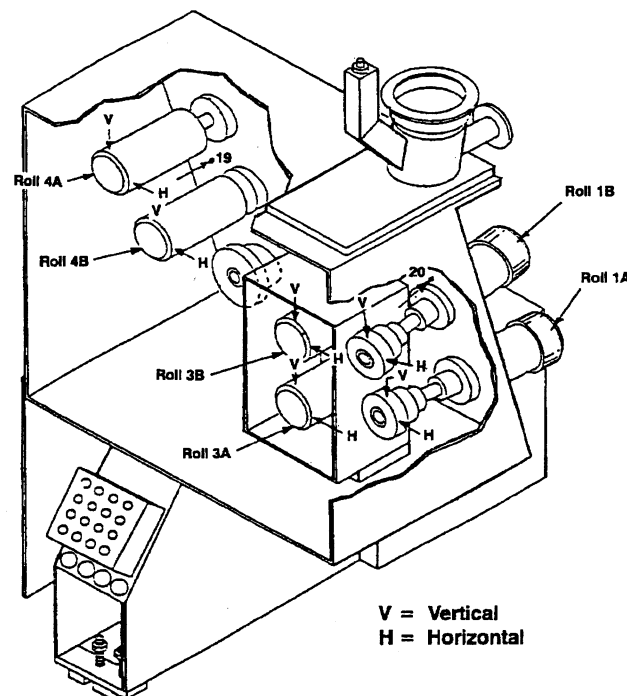


FIGURE 10
GODET ROLL PANEL NATURAL FREQUENCY MEASUREMENT POSITIONS

Please refer to the cascade diagram of Figure 11 which shows results of a controlled runup test for Godet Roll 3A using an AC motor controlled by an inverter. Several interesting things appear on the waterfall of Figure 11. First, one can see the resonances at about 2730 CPM, 4250 CPM, 5978 CPM and then in an entire region from approximately 6400 to 7100 CPM. This plot also shows something else very unusual which began to occur at a speed of approximately 6300 RPM. Note the subsynchronous frequencies that began to appear at this point and remained up to a speed of approximately 6800 RPM. While one set of subsynchronous frequencies was increasing in frequency with an increase in speed, another subsynchronous set was actually decreasing with increasing rotor speed. Later, this was determined to be an instability of the Godet Roll bearings. After much study, it was determined that the bearings had improper preload (exerted by wavy springs) and also improper interference fit. Therefore, when the rotor began going into the resonant amplification zone between 6400 and 7100 RPM, this bearing instability would occur.

Figure 12 shows the waterfall spectrum during a controlled runup test up to 10,800 RPM after a series of modifications were made to the Godet Roll. Note the complete elimination of the structural resonance problems and also note that the roll fundamental natural frequency was successfully moved from about 6000 up to 8520 RPM. In addition, even though the rotor had the 8520 RPM fundamental natural frequency, it now suffered much lower amplitudes of only .260 in/sec even when resonant compared to a level of .953 in/sec at the rotor fundamental before. This was due to a series of modifications including changes in shaft diameters, material changes and both damping and isolation treatments.

Interestingly, note that none of the bearing instability previously encountered at speeds of only 6100 RPM occurred again until a speed of about 10,800 RPM was attained. Strangely, at this point, a subsynchronous frequency at exactly 2X the bearing cage defect frequency would occur almost every time this was attempted (cage frequency was .411 times RPM). At this time, the predominant vibration would not be at 1X rotor RPM, but at the 2X bearing cage frequency. Reasons for instability at this particular frequency were never completely understood. However, as long as roll speeds remained below 10,000 RPM, this bearing instability would not occur.

The most important success with regard to the client was that they now had a machine which could successfully run not only up to the desired minimum of 6000 RPM, but now up to as high as 8000 RPM with amplitudes remaining below .200 in/sec. In addition, the vendor now had a machine which could enter a whole new market up to speeds of 8000 RPM which corresponded to production speeds of about 4200 MPM.

Note from the cascade diagrams in this section that during a runup test, all peaks which are harmonic or subharmonic orders of running speed closely approximate straight lines. As the machine increases in speed, the slope of the "lines" are skewed to the right, the slope depending on the running speed order (or harmonic). On the other hand, during a coastdown, the slope of the "lines" are skewed to the left as the machine coasts to a stop. In both cases, spectral peaks which are not harmonically related (nonsynchronous or asynchronous) or those that represent instability, move in random directions on the plot.

This section enabled the analyst to see that transient events, rubs, oil whirl, oil whip, resonance and bearing instability can actually be "seen" with the use of cascade plots. The behavioral patterns as well as frequencies could be identified as they interacted with others. This is truly an important diagnostic "tool in the vibration analyst's tool box".

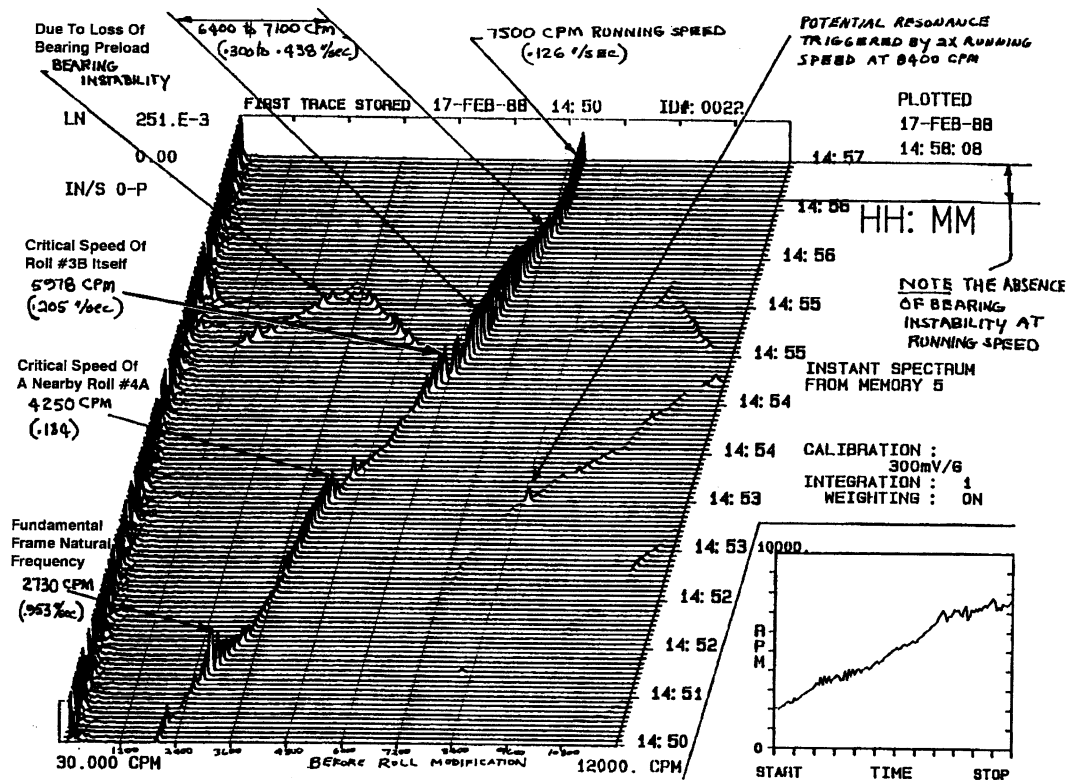


FIGURE 11
CONTROLLED RUNUP TEST FOR GODET ROLL 3A

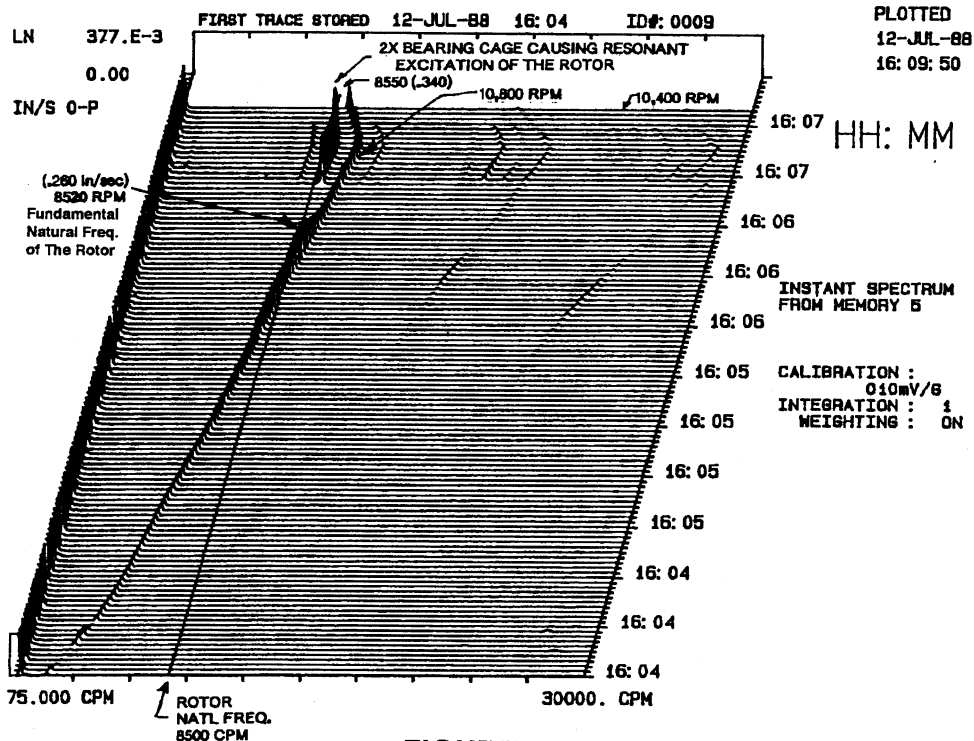


FIGURE 12
CONTROLLED RUNUP TEST AFTER MODIFICATIONS TO GODET ROLL 3A

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2. Bently, Donald E.; President and Chairman, Bently Nevada, Inc.; Minden, NV; "Rubs Research - Studies Reveal Physical Phenomena of Rotor Rubs"; Orbit, Volume 4, No. 3; pages 3-5; October, 1983.
3. Bently, Donald E.; Zimmer, Siegfried; Palmatier, George E. and Muszynska, Agnes; "Interpreting Vibration Information From Rotating Machinery"; Sound and Vibration Magazine; Volume 20, No. 2; pages 14-23; February, 1986.

